

Andrew Russell

ENGINEER & TECHNICAL LEADER · SECURITY, DATA & PRODUCT

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Summary

I build technical systems and lead the work around them. My background spans cryptography research, security and privacy engineering at Google and Microsoft, and data, product, and business leadership at Apollo Agriculture. I enjoy understanding difficult systems, testing ideas against evidence, and taking responsibility for outcomes. My next role should combine technical work with a tangible positive impact, particularly making AI safer and more useful. Currently on sabbatical, including a completed three-month solo motorbike expedition from Nairobi to Namibia, with independent software projects along the way. Available to start beginning December 2026.

Work Experience

Sabbatical

Travel & independent projects

Jan 2026 – Present

- Completed a three-month solo motorbike expedition from Nairobi to Namibia (Jun – Sep 2026).
- Designed, built, and shipped **Costanza**, an AI agent managing a charitable treasury, with auction-based execution and contract-enforced action limits. See Selected Works below.
- Built and shipped **It Was Never You**, an iOS art app about AI and personal memory, live on the App Store.

Apollo Agriculture

Netherlands / Kenya / Zambia

Senior Product Manager

May 2024 – Dec 2025

- Appointed by C-suite to lead Zambia's \$10M revenue business unit; accountable for commercial performance, product, and operations across sales, credit, and engineering.
- Authored two company-wide retrospectives following major customer contraction, combining market research, customer/staff feedback, and internal operational analysis; findings drove pivots in business model and multi-year growth strategy.
- Built experimental products, including a wholesaling initiative to reduce balance sheet risk from multi-million-dollar fertilizer holdings.

Software Engineer → Tech Lead → Head of Data

Apr 2021 – Mar 2024

- Built and scaled Apollo's core data infrastructure in SQL and Python. Led migration of legacy analytics to a modern stack, improving query speeds by 97% and cutting costs by 60%.
- Led cross-functional analytics supporting operations across multiple countries.
- Promoted to Head of Data; managed a team of 8 engineers and analysts across Europe and Africa.

Google

San Francisco, CA

Software Engineer

Aug 2019 – Mar 2021

- Built a system monitoring petabytes of user data for daily privacy compliance across Google's product ecosystem.

Software Engineer, PhD Intern

Summer 2018

- Designed a threat model for a privacy-preserving version of Gmail using secure execution environments; ported part of Gmail's email-processing backend into the system.

Microsoft

Redmond, WA

Software Engineer

Aug 2016 – Aug 2017

- Worked on secure sign-in and sign-out flows for Windows 10 as part of the Windows Security team (Windows Hello).

Selected Works

Costanza: The Human Fund (2026)

thehumanfund.ai · github

Built an AI agent that manages a charitable treasury on Base and pays external operators to run inference through a reverse auction. Combined Intel TDX hardware attestation with contract-enforced action and spending limits. Wrote a formal security and liveness model with explicit trust assumptions, proof sketches, and game-theoretic analysis. [Technical whitepaper](#).

It Was Never You (2026)

[itwasneveryou.app](#) · [App Store](#)

Built and shipped an iOS app as a satirical art piece that uses generative AI to replace a selected person in photographs. Explores how AI image manipulation can undermine personal identity and history as well as the public record. Available on the App Store.

Publications & Writing

Risky Traitor Tracing and New Negative Differential Privacy Results

[CRYPTO 2018](#)

with Rishab Goyal, Venkata Koppula, and Brent Waters

I built an AI agent I can't turn off. Now it won't listen to me.

[Substack, Jul 2026](#)

Reflections on operating Costanza: agent behavior, human influence, and control

Education

The University of Texas at Austin

[Austin, TX](#)

M.S. in Computer Science & Ph.D. coursework

2017 – 2019

Cryptography (obfuscation, homomorphic/functional encryption). Advised by Brent Waters.

Pomona College

[Claremont, CA](#)

B.A. in Mathematics and Computer Science

2012 – 2016

Thesis: Elliptic Curve Pairings in Cryptography. Summer research fellowships at Stanford (2014) and the University of Bristol (2015).

Fun Stuff

Division I club ultimate frisbee athlete on a team that competed at nationals; two-time Dutch national champion. Ultramarathoner and advanced-certified freediver (35m). Eagle Scout; hiked and backpacked across 25+ countries. Violinist for 15+ years.